

ACME Inc. — Medallion Data Architecture

EDW Layer Design | Microsoft Fabric Lakehouse | June 2026

Architecture Overview

BRONZE — Raw	SILVER — Trusted	GOLD — BI-Ready
PURPOSE: Exact, immutable mirror of all source files. Full auditability. No transformations applied.	PURPOSE: Single source of truth. Typed, cleaned, enriched, and validated. Business logic excluded.	PURPOSE: Star schema optimized for BI. Two fact tables share conformed dimensions (Constellation Schema).
SOURCES: 11 source files ingested as 12 Delta tables (CSV, XML, Excel). Shipments merged from 3 files.	TRANSFORMATIONS: Dates parsed (M/d/yyyy). Geographic enrichment (Continent, SubRegion, ISO2). Known errors corrected. Unknown Members injected for referential integrity.	FACTS: fact_sales_items (17,032 rows, line grain), fact_orders (6,571 rows, order grain), fact_budget (40 rows, employee x year)
KEY FEATURE: _ingested_at timestamp on every table. Schema inferred at read time.	GOVERNANCE: Invalid records quarantined in bronze_quarantine (not silently dropped). All fixes traceable.	DIMENSIONS: dim_date (2007–2021), dim_customer, dim_product, dim_employee, dim_shipper
TABLES (12): bronze_orders, bronze_order_details, bronze_shipments, bronze_customers, bronze_products, bronze_categories, bronze_divisions, bronze_shippers, bronze_suppliers, bronze_employees, bronze_offices, bronze_budget	TABLES (12): silver_ prefix for all source tables. Plus bronze_quarantine for rejected records.	METRICS: SalesAmount, GrossMargin, GrossMarginPct, TotalOrderValue, BudgetMonthly, DeliveryDays
		AUDIT: gold_dq_issues — all DQ checks, BI-ready

Pipeline Workflow



Orchestration: Fabric Data Pipeline — notebooks execute in dependency chain. Each step runs only if previous completed successfully.

Implications of This Architecture

STRENGTHS	CONSIDERATIONS & NEXT STEPS
Separation of concerns: Raw data preserved independently of transformations — Bronze never overwritten.	Full-reload pipeline: Production requires incremental loads (MERGE/UPSERT) for performance and idempotency.
Full auditability: Any issue traced back to Bronze without re-ingestion. Every fix is documented.	GrossMargin accuracy: Uses current UnitCost — SCD Type 2 on Products needed for historical accuracy.
Scalable: Each layer evolves independently without breaking downstream consumers.	Shipment SLA: Blocked by legacy date data (2007–2012 vs 2016–2021). Engage logistics team.
Governance: Quarantine framework makes DQ issues explicit and trackable via gold_dq_issues.	DivisionID=2 violation: PK conflict requires source correction before go-live.
BI-ready: Direct Lake on Delta tables eliminates import latency in Power BI.	Unknown Members: Shippers 4 & 5, Employee -1 are interim — register in source systems.